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Switching Techniques

ET3003 Computer Networks

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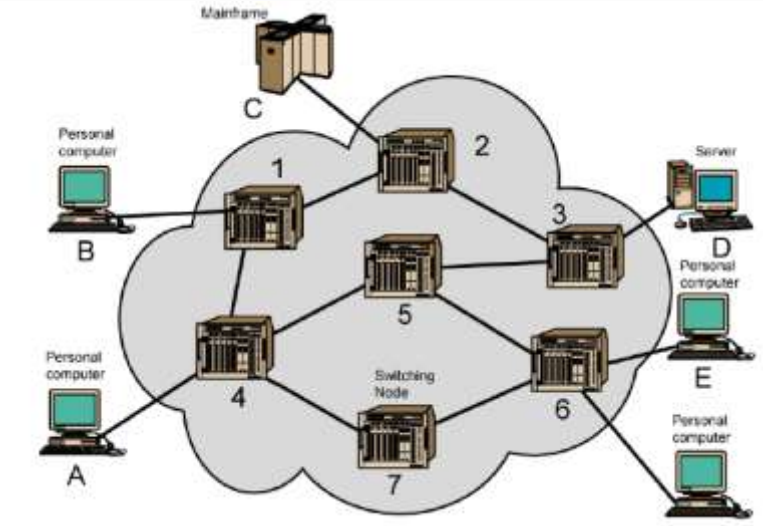
Switching Networks

→ A collection of nodes and connections is a communications network

→ Data is routed by being switched from node to node



How to Switch the Data?

- Circuit switching
- Packet switching
  - Datagram
  - Virtual circuit

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Circuit switching

- Communication via circuit switching involves three phases:
  - Circuit Establishment → connection oriented
  - Data Transfer
  - Circuit Disconnect
- Circuit → *Dedicated* communication path between two stations
- Developed for voice traffic (phone; Public Switched Telephone Network (PSTN))
  - Delay (and delay variation (jitter)) is guaranteed as required by voice traffic

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- Circuit switching is inefficient
  - Channel capacity dedicated for duration of connection → If no data, capacity wasted
- Information is transmitted at a fixed data rate with no delay (except for the propagation delay)

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Packet Switching

- Station breaks long message into packets
- Packets sent one at a time to the network
- Packets handled in two ways
  - Datagram
  - Virtual circuit
- *Network resources are shared*

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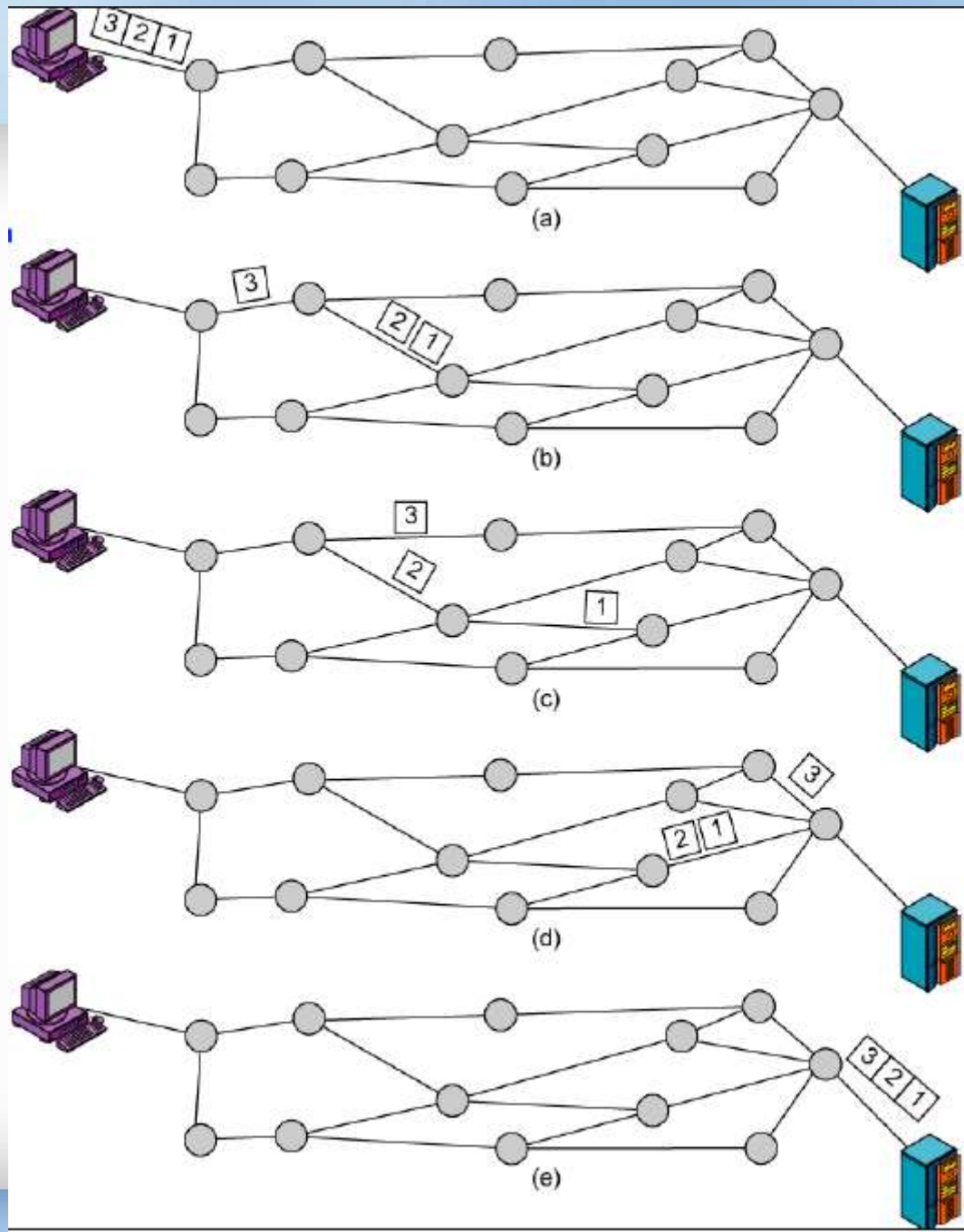
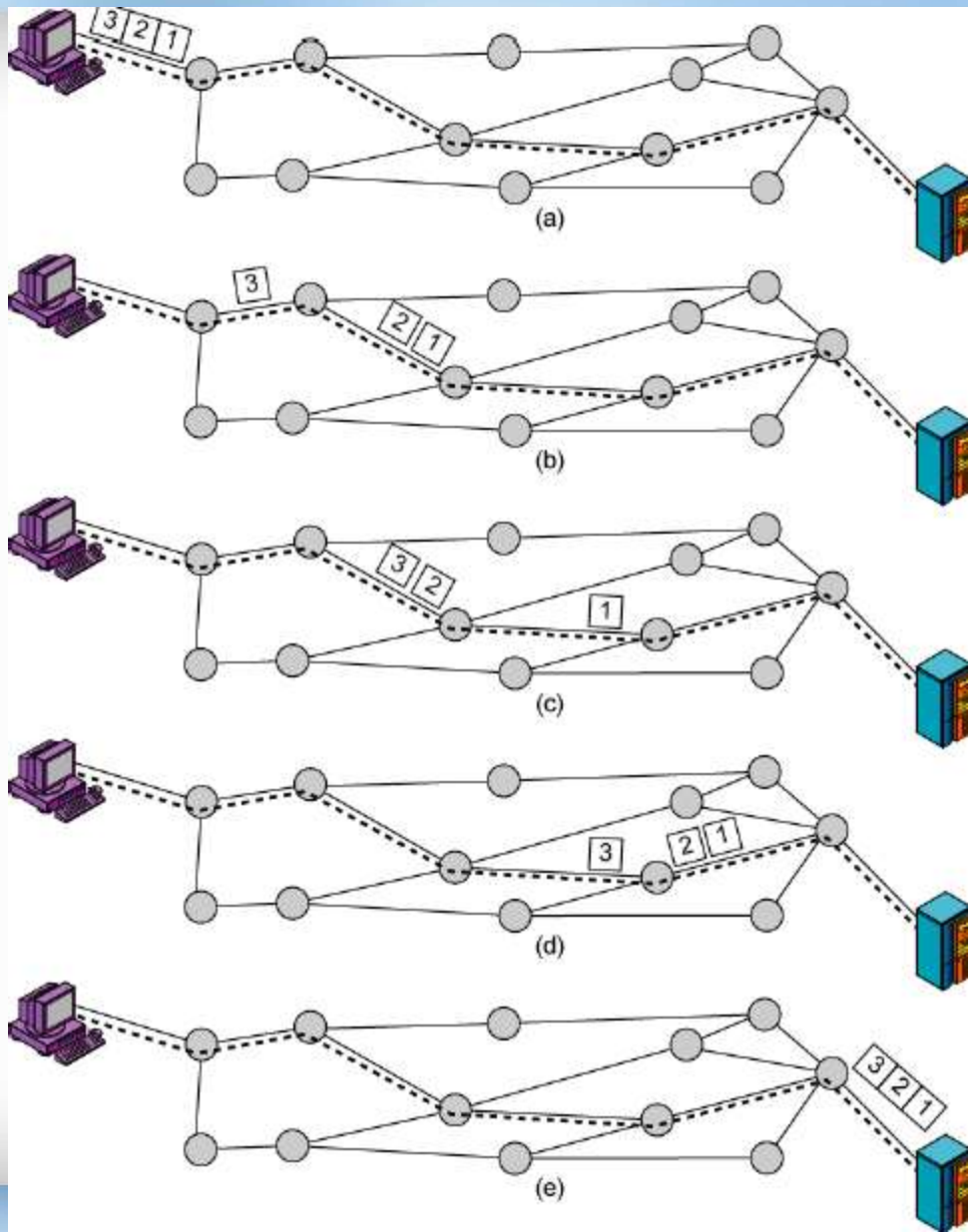
Datagram

- *There's no preplanned route established* before any packets sent → connectionless
- Each packet treated independently
- Packets can take any practical route
- Packets may arrive out of order
- Packets may go missing
- Up to receiver to re-order packets and recover from missing packets



Virtual Circuit

- *A preplaned route is established* before any packets are sent → connection oriented
  - Call request and call accept packets establish connection (handshake)
- Once route is established, all the packets follow same route
- Each packet contains a Virtual Circuit Identifier (VCI)
- No routing decisions required for each packet
- Clear request to drop circuit
- *Not a dedicated path*



Circuit switch vs virtual circuit (an analogy)

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